Module 5 – Core PHP

1. Discuss the structure of a PHP script and how to embed PHP in HTML.

Ans :

1) Structure of a PHP Script and How to Embed PHP in HTML

English

A PHP script is a file with the .php extension that contains PHP code. The PHP code is executed on the server, and the result is sent to the browser as HTML.

Basic structure of a PHP script:

```
<?php
   // PHP statements
   echo "Hello, World!";
?>
```

Main parts:

- <?php → starts the PHP code block
- ?> → ends the PHP code block
- echo → prints output
- ; → ends each statement

Example:

```
<?php
   $name = "Bharat";
   echo "Welcome " . $name; // that is using the concating
?>
```

Embedding PHP in HTML

PHP can be written inside HTML to make web pages dynamic.

Example:

```
<!DOCTYPE html>
<html>
<head>
   <title>PHP Example</title>
</head>
```

```
<body>

<h1>My Web Page</h1>

<?php
   echo "<p>Hello from PHP!</p>";
?>

</body>
</html>
```

How it works:

- HTML gives the page structure.
- PHP generates dynamic content.
- The server processes PHP first, then sends the final HTML to the browser.

Another example:

```
<p>
<?php
   echo "Today is " . date("Y-m-d");
?>
</p>
```

2) What are the rules for naming variables in PHP?

Ans :

Rules for Naming Variables in PHP

English

1. A PHP variable must start with $.
2. After $, the first character must be a letter or underscore (_).
3. It cannot start with a number.
4. It can contain only letters, numbers, and underscores.
5. No spaces are allowed.
6. Variable names are case-sensitive.

7. Do not use PHP reserved keywords as variable names.

Examples

```
$name = "Bharat";
$_age = 20;
$user1 = "OK";
```

Invalid examples

```
$1name = "Wrong";
$user name = "Wrong"
```

3) Explain the concept of variables in PHP and their scope.

Ans :

A variable in PHP is a name used to store data such as numbers, text, arrays, or objects.
It acts like a container where values can be kept and changed during program execution.

In PHP, variables always start with the $ symbol.

Example:

```
<?php
$name = "Bharat";
$age = 20;
?>
```

Here:

- $name stores the text "Bharat"
- $age stores the number 20

4) What are super global variables in PHP? List at least five super global arrays and their use.

Ans :

Superglobal variables in PHP are built-in variables that are available in all scopes of a script.
You do not need to declare them with global inside a function. PHP automatically makes them accessible everywhere.

1. $_GET
   Used to collect data sent through the URL query string.
   Example: form data submitted by the GET method.
2. $_POST
   Used to collect form data sent through the HTTP POST method.
   Example: login forms, registration forms.
3. $_REQUEST
   Used to access data from $_GET, $_POST, and $_COOKIE.
   It is useful when you want to read request data from multiple sources.
4. $_SERVER
   Contains server and execution environment information.
   Example: server name, script name, request method, IP address.
5. $_SESSION
   Stores session data for a user across multiple pages.
   Example: keeping a user logged in.
6. $_COOKIE
   Used to access values stored in browser cookies.
   Example: remembering user preferences.
7. $_FILES
   Used to handle file uploads through forms.
   Example: uploading an image or document.

5) Explain how conditional statements work in PHP.

Ans :

Conditional statements in PHP are used to make decisions. They check whether a condition is true or false, and then run different blocks of code based on the result.

Why they are used

They help the program respond differently in different situations.

For example:

- if age is 18 or more, allow voting
- if marks are above 40, pass the student
- if a number is even, print “Even”

- Main conditional statements in PHP
- 1) `if` statement
- Runs code only when the condition is true.
- ```
  <?php
  $age = 20;

  if ($age >= 18) {
      echo "You are an adult";
  }
  ?>
  ```
- Here, the message is shown only because `$age >= 18` is true.

- 2) `if...else` statement
- Chooses between two blocks of code.
- ```
  <?php
  $age = 16;

  if ($age >= 18) {
      echo "You can vote";
  } else {
      echo "You cannot vote";
  }
  ?>
  ```
- If the condition is true, the first block runs.
  If it is false, the `else` block runs.

- 3) `if...elseif...else`
- Used when there are multiple conditions.
- ```
  <?php
  $marks = 75;

  if ($marks >= 90) {
      echo "Grade A";
  } elseif ($marks >= 60) {
      echo "Grade B";
  } else {
      echo "Grade C";
  }
  ?>
  ```

6) Discuss the difference between for loop, foreach loop, and do-while loop in PHP.

Ans :

In PHP, for, foreach, and do-while loops are used to repeat a block of code, but they work in different ways.

1) for loop

A for loop is used when the number of repetitions is known or when you want to control the loop with a counter.

Syntax:
  ```

```
for (initialization; condition; increment) {
   // code
}
```

Example:

```
<?php
for ($i = 1; $i <= 5; $i++) {
   echo $i . " ";
}
?>
```

2) foreach loop

A foreach loop is used mainly to loop through arrays. It is the easiest loop when you want to read every element in an array.

Syntax:

```
foreach ($array as $value) {
   // code
}
```

Example:

```
<?php
$colors = array("Red", "Green", "Blue");

foreach ($colors as $color) {
   echo $color . " ";
}
?>
```

3) do-while loop

A do-while loop runs the code at least once, because the condition is checked after the loop body.

Syntax:

```
do {
   // code
} while (condition);
```

Example:

```
<?php
$i = 1;

do {
   echo $i . " ";
   $i++;
} while ($i <= 5);
?>
```

7) Define arrays in PHP. What are the different types of arrays?

Ans :

PHP, an array is a variable that can store multiple values in one variable.

Example:

```
$fruits = array("Apple", "Banana", "Mango");
```

Types of arrays in PHP

1. Indexed Array
   Uses numeric indexes like 0, 1, 2.

```
$colors = array("Red", "Green", "Blue");
echo $colors[1]; // Green
```

2. Associative Array
   Uses named keys instead of numbers.

```
$student = array("name" => "Bharat", "age" => 20);
echo $student["name"]; // Bharat
```

3. Multidimensional Array
   An array inside another array.

```
$students = array(
   array("Bharat", 20),
   array("Rahul", 22)
);
echo $students[0][0]; // Bharat
```

8) What is the header function in PHP and how is it used?

Ans :

In PHP, the header() function is used to send raw HTTP headers to the browser before any output is sent.

It is commonly used for:

- Redirection
- Setting content type
- Sending cache control headers
- Triggering file downloads

Common use: redirect to another page

```
<?php
header("Location: index.php");
exit;
?>
```

Important points

- header() must be called before any HTML or echo output.
- After redirection, use exit; so the script stops immediately.

9) Explain the difference between include and require in PHP.

Ans :

PHP, include and require both add one file into another file.

The difference is what happens when the file is missing:

include
If the file is not found, PHP gives a warning and the script keeps running.

```
include "header.php";
echo "Hello";
```

require
If the file is not found, PHP gives a fatal error and the script stops immediately.

```
require "config.php";
echo "Hello";
```

Simple meaning

- include = optional file

- require = compulsory file

Extra versions

- include_once → file only one time
- require_once → file only one time

Use

- Use include for header, footer, menu
- Use require for database connection, config files

10) Explain what PHP expressions are and give examples of arithmetic and logical operations.

Ans :

Arithmetic operations

Arithmetic expressions are used for math calculations.

Common operators:

- + addition
- - subtraction
- * multiplication
- / division
- % remainder

Example:

```
<?php
$a = 10;
$b = 3;

echo $a + $b; // 13
echo $a - $b; // 7
echo $a * $b; // 30
echo $a / $b; // 3.333...
echo $a % $b; // 1
?>
```

Logical operations

Logical expressions are used to compare conditions and return true or false.

Common operators:

- && means AND
- || means OR
- ! means NOT

Example:

```
<?php
$a = 10;
$b = 5;

var_dump($a > 5 && $b < 10); // true
var_dump($a < 5 || $b == 5);  // true
var_dump(!($a == 10));       // false
?>
```

Easy difference

- Arithmetic expressions → do calculations
- Logical expressions → check conditions